Counterexamples for interacting particle systems

Overview

github.com/jonaskoeppl/counterexamples-ips

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This repository collects counterexamples to plausible statements about interacting particle systems in the setting of Liggett's books, found by humans or with the help of AI systems. Each entry below states the question in one paragraph and links to the full write-up. The status *plausible* means that the argument looks right but has not yet been checked in full by a human.

0001 One reversible measure, but a non-reversible stationary measure: exclusion processes

Status: *plausible*

If an interacting particle system has at least one reversible measure, is every stationary measure reversible? For stochastic Ising models this is the Holley–Stroock question: true in $d \leq 2$, open in $d \geq 3$. For general systems the answer is no. For nearest-neighbour asymmetric simple exclusion on $\mathbb{Z}$, the blocking measures are reversible, while the Bernoulli product measures are stationary but carry a non-zero current and are therefore not reversible. Any positive statement must exclude conservative dynamics of this kind.

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0002 Symmetric exclusion on two half-spaces of $\mathbb{Z}^3$: a non-reversible stationary measure

Status: *plausible*

For exchange dynamics on a connected graph of bounded degree, is every stationary measure with finite energy reversible? On recurrent graphs, in particular $\mathbb{Z}^d$ with $d \leq 2$, the answer is yes, by the conservative analogue of Holley–Stroock. On transient graphs it is no. Join two half-spaces of $\mathbb{Z}^3$ by a single bond: symmetric simple exclusion on this graph has a stationary measure with finite energy whose density tends to different limits $\rho_- < \rho_+$ in the two half-spaces, and the resulting current through the bridge rules out reversibility. Found by ChatGPT Astra-6.

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0003 Directed noise can destroy ergodicity

Status: *plausible*

Bérard, Dembin and Marêché (arXiv:2510.11424) prove that a monotone ergodic system with **0** absorbing becomes exponentially ergodic when each site is additionally set to 0 at rate ε. Can the absorbing state be dropped? No, not even for plain ergodicity. For the Ising-type dynamics with constant rate $+1 \to -1$ and rate $(1+\varepsilon)e^{2\beta S_x}$ for $-1 \to +1$ (an external field $h = \frac{1}{2}\log(1+\varepsilon)$), adding the noise is the same as removing the field. For $d \geq 2$ and $\beta > \beta_c$ the original dynamics is ergodic, while the perturbed one has both the plus and the minus Gibbs state as reversible measures.

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0004 Positive rates do not force stationary measures to be quasilocal (on a tree)

Status: *plausible*

Consider an interacting particle system with finite local state space and finite-range rates that are uniformly bounded away from zero, with no forbidden transitions (so that conservation laws are ruled out). Is every stationary measure quasilocal? The question is inspired by elliptic regularity: stationary measures solve $\mathscr{L}^*\mu = 0$, positive rates play the role of ellipticity and quasilocality that of regularity. On $\mathbb{Z}^d$ it is open. On the 3-regular tree the answer is no: an explicit nearest-neighbour pair-update dynamics with three states and all rates at least $1.9 \cdot 10^{-4}$ has two stationary product measures with different densities, and their mixture is stationary but its conditional probabilities depend on the tail, so it is not quasilocal. Branching is essential: on $\mathbb{Z}$ two such product measures cannot coexist. Found by ChatGPT Astra-6.

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